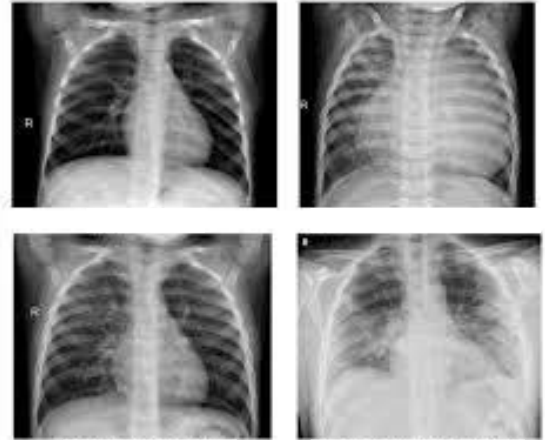


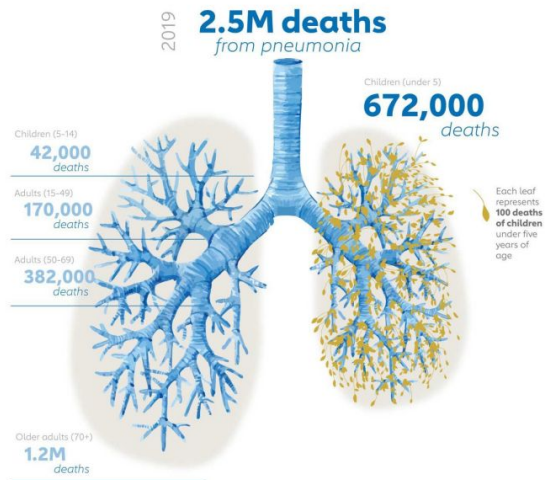
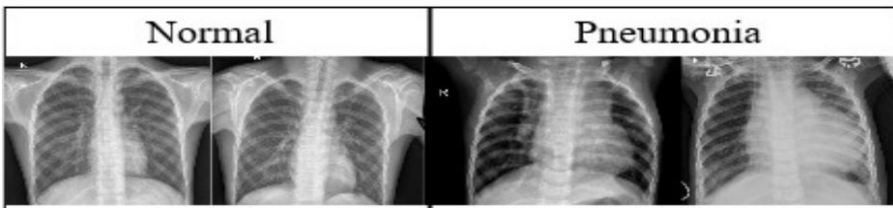
# Pneumonia Classification Using CXR Images with Deep Learning and Model Explainability

Hritesh Rajanagan, Steven Huynh, Ethan Kittell



# Problem Overview & Goals

- **Objective:** develop accurate, efficient, and reliable models for Pneumonia classification
- **Why?**
  - High mortality rate (children & elderly)
  - Bacterial vs. Viral
  - misclassification



# Dataset Summary

- **Dataset** - [Chest X-Rays: bacterial / viral / normal](#)
- ~5000 images on training Folder
  - 80/10/10 split
  - 3 classes
    - 0: no pneumonia
    - 1: bacterial pneumonia
    - 2: viral pneumonia
- Labeled metadata
- Problems with Dataset:
  - Test Folder missing labels
  - Even smaller Dataset



KOSTAS DIAMANTARAS · UPDATED 4 YEARS AGO



## Chest Xrays: bacterial / viral pneumonia / normal

Chest Xray classification

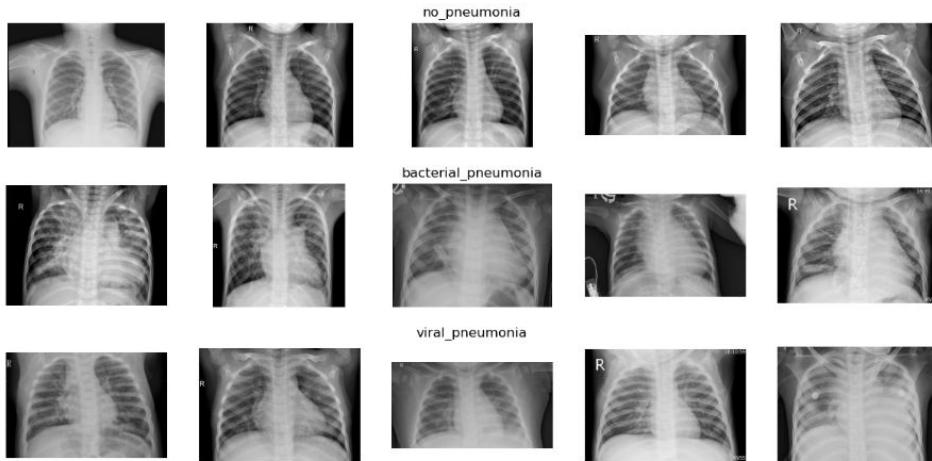


Data Card

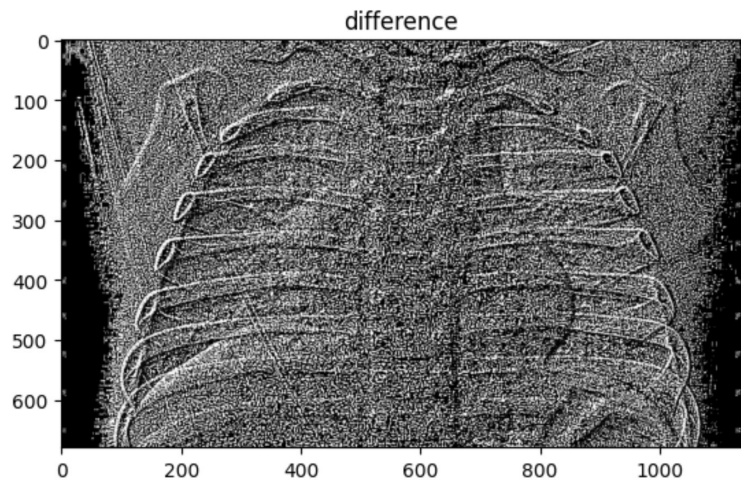
Code (4)

Discussion (0)

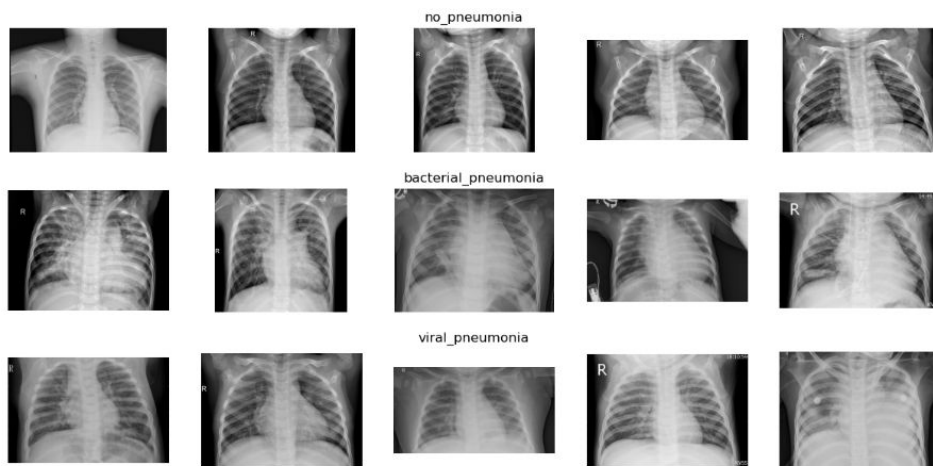
Suggestions (0)



# Summary of EDA

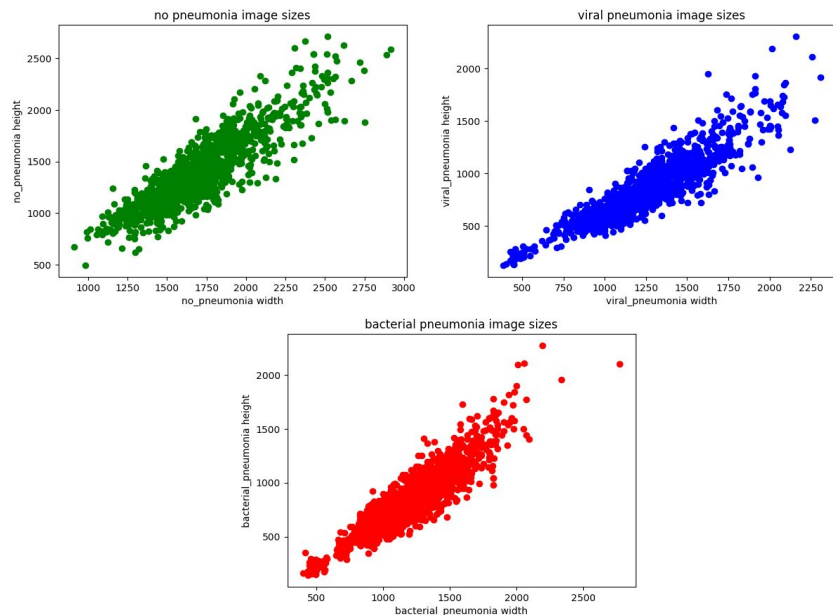


Noisy Data

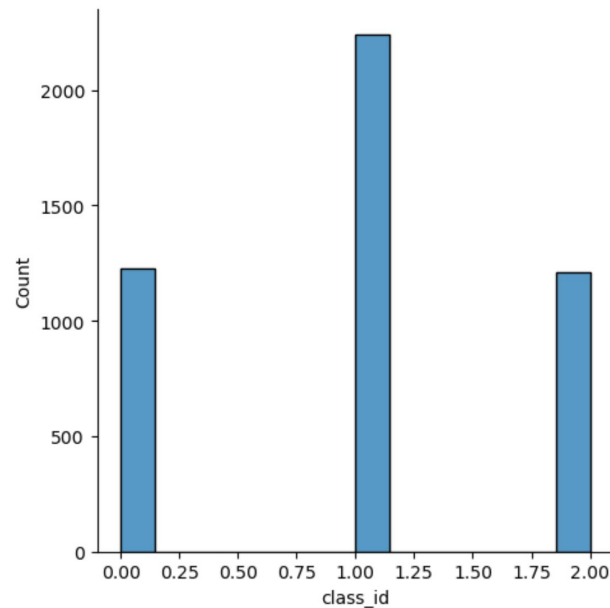


Easily Misclassified Data

# Summary of EDA cont'd



Non-uniform Data



Imbalance Data

# Influence of EDA

```
sorted_classes = sorted(class_weights.keys())
weights_list = [class_weights[c] for c in sorted_classes]
weights_tensor = torch.FloatTensor(weights_list)

print(sorted_classes)

[7] ✓ 0.0s

... 1 : 0.5482573726541554
     2 : 1.0165700082850042
     0 : 1.0
     [np.int64(0), np.int64(1), np.int64(2)]

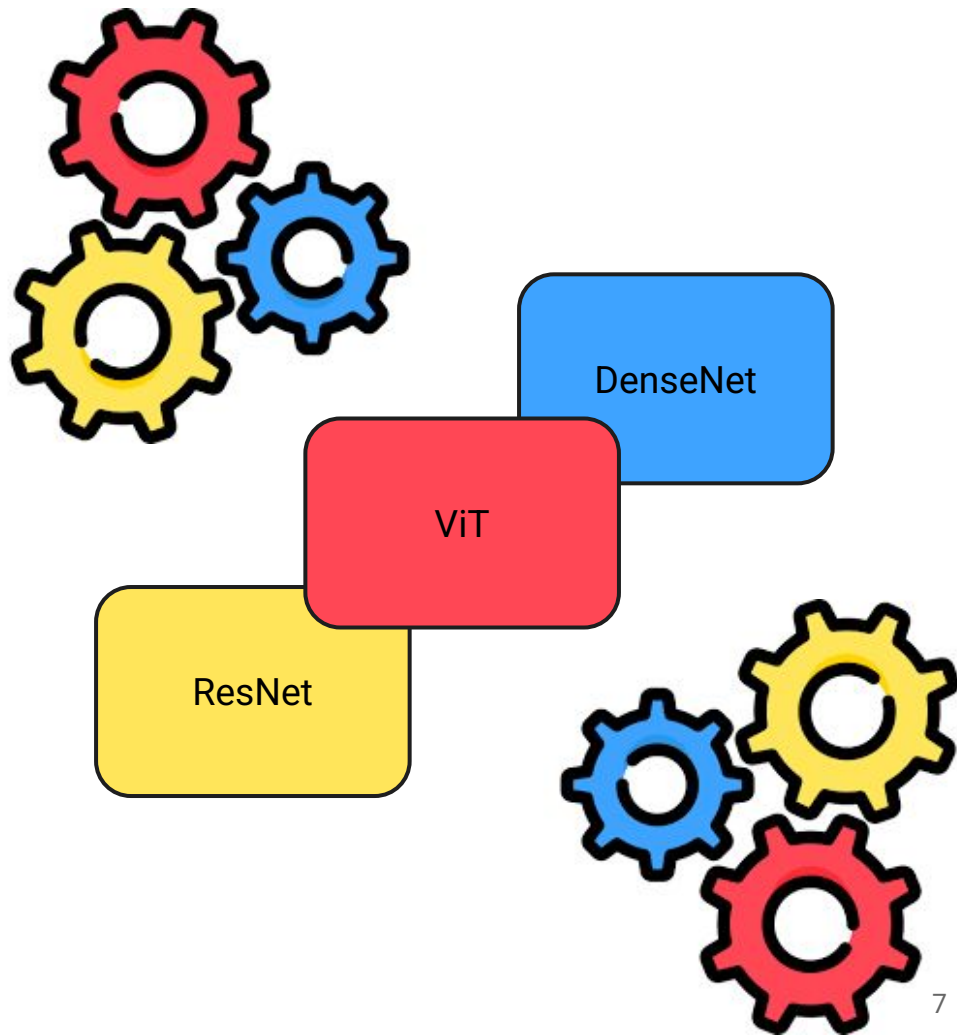
transform = transforms.Compose([
    transforms.RandomResizedCrop(224),
    transforms.RandomRotation(30),
    transforms.RandomVerticalFlip(),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.5]*3, std=[0.5]*3)
])

transform_val = transforms.Compose([
    transforms.Resize(256),           # resize the shorter side to 256
    transforms.CenterCrop(224),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.5]*3, std=[0.5]*3)
])
```

- Hints at expected challenges
  - Viral vs. normal
- Guides transformations
  - Resize
  - No denoising
  - Random transformations
- Alludes to rebalancing measures
  - Median frequency balancing

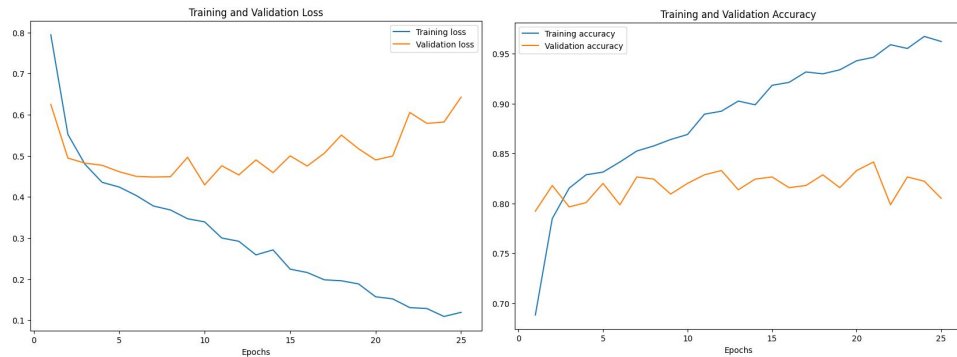
# Key Experiments

- Models Architectures:
  - ResNet-18\*\*
  - DenseNet-121
  - Data efficient transformer
- Transformations
- Hyperparameters
  - Batch sizes
  - Learning rate
  - Dropout
  - Optimizer

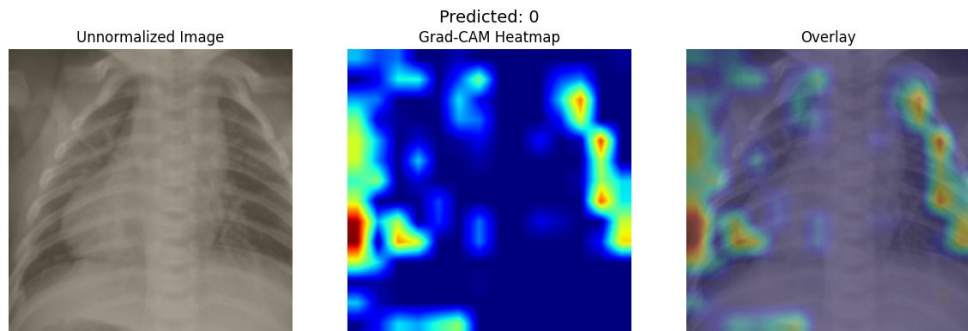


# Experimental Evaluation: Metrics

- Loss & Accuracy
  - Training & validation
- Confusion Matrix
  - Precision & recall
- Grad-CAM
  - Reliability

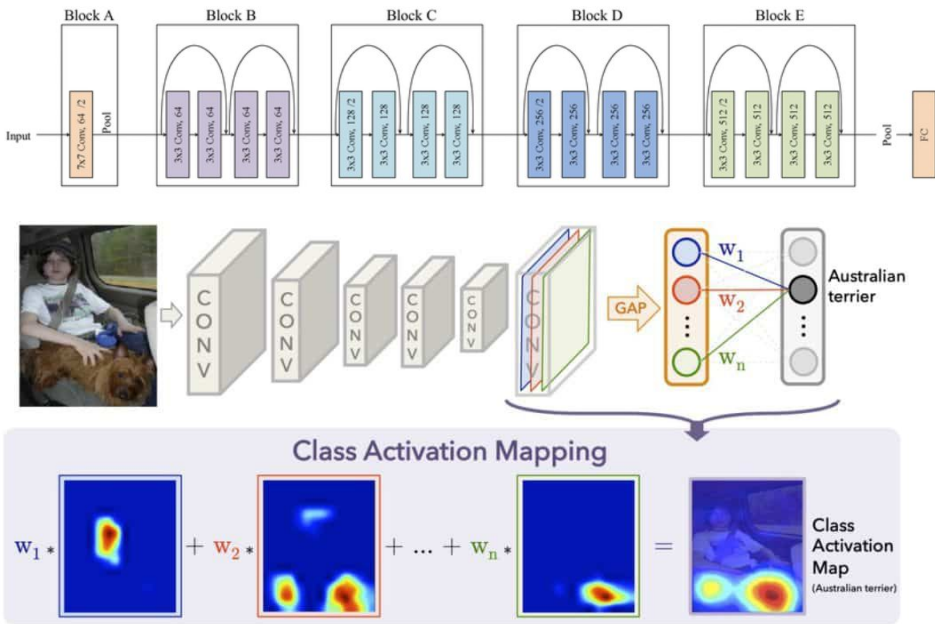


|                 |          | True Class     |                |
|-----------------|----------|----------------|----------------|
|                 |          | Positive       | Negative       |
| Predicted Class | Positive | True Positive  | False Positive |
|                 | Negative | False Negative | True Negative  |

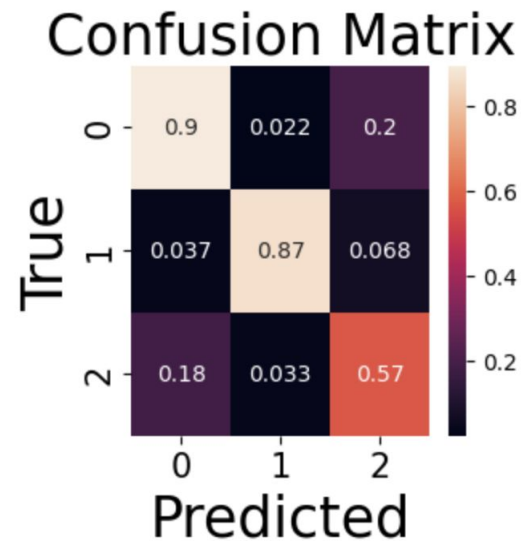
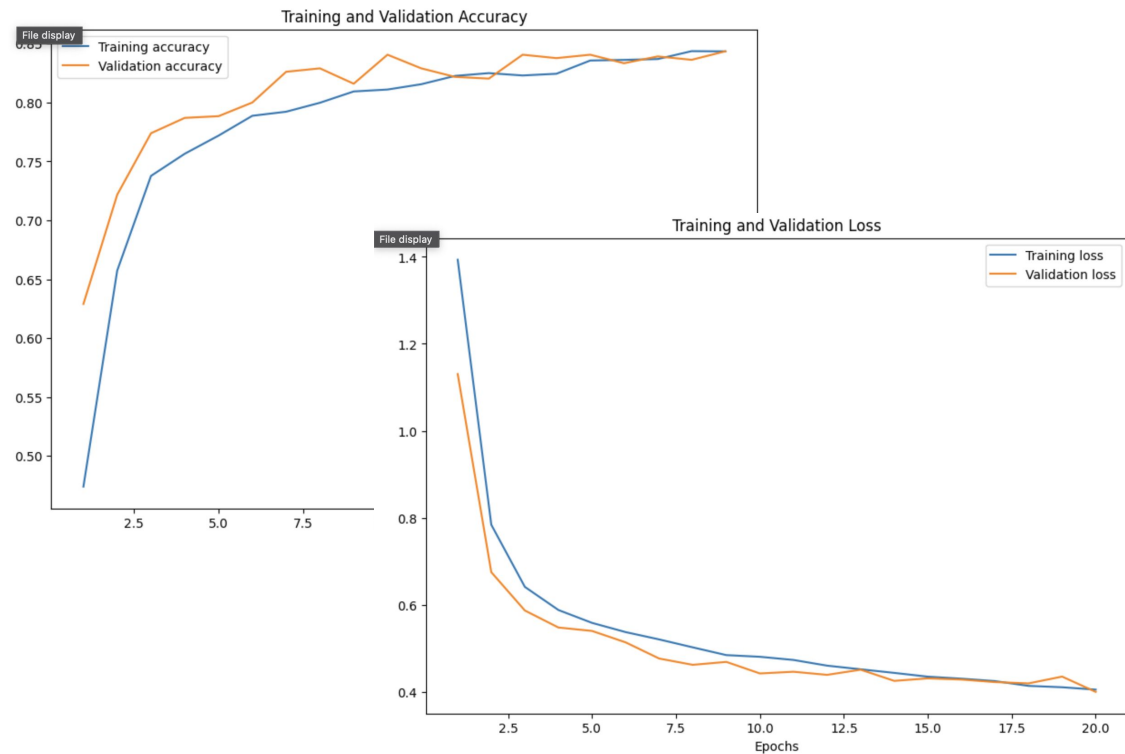


# Key Experiments: ResNet-18

- Note: Switched from ResNet50
  - Due to loss of test dataset
- Baseline Model
  - Residual Network
  - 18 layers
- 20 Epoch Ran
- Heuristics
  - loss/accuracy
  - Confusion matrix
  - Grad Cam

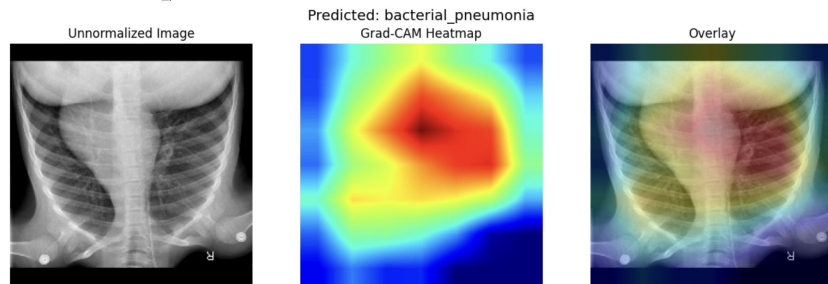


# Key Experiments: ResNet-18 Results

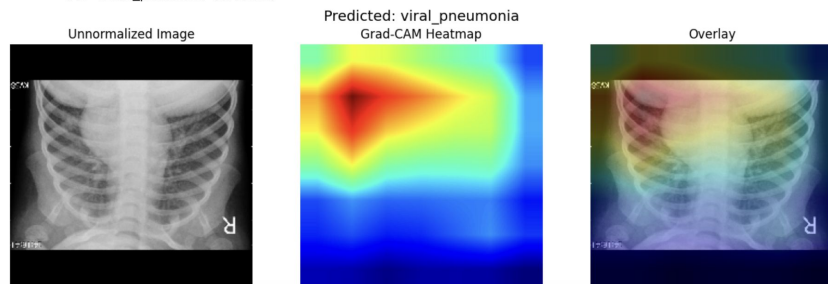


# Key Experiments: ResNet-18 Results (cont)

True Class: bacterial\_pneumonia  
Generating Grad-CAM @backbone.7.1.bn2  
#0: bacterial\_pneumonia (0.02041)



True Class: viral\_pneumonia  
Generating Grad-CAM @backbone.7.1.bn2  
#0: viral\_pneumonia (0.00038)

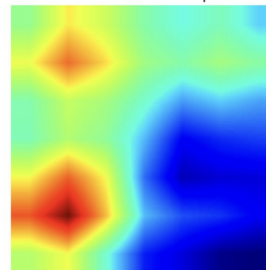


True Class: no pneumonia

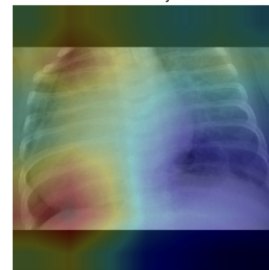
Unnormalized Image



Predicted: no\_pneumonia  
Grad-CAM Heatmap

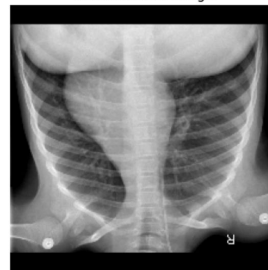


Overlay

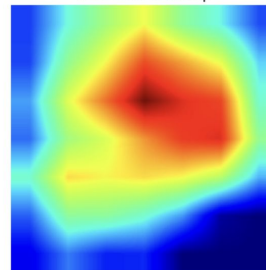


Generating Grad-CAM @backbone.7.1.bn2  
#0: bacterial\_pneumonia (0.02041)

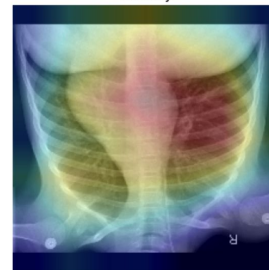
Unnormalized Image



Predicted: bacterial\_pneumonia  
Grad-CAM Heatmap



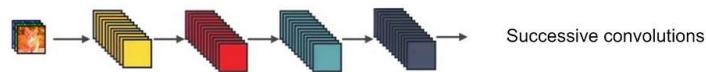
Overlay



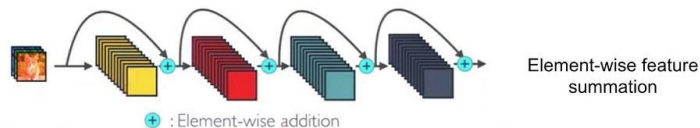
# Key Experiments: DenseNet-121

- Baseline Model
  - DenseNet
  - 121 layers
- Architecture:
  - Dense Blocks
  - Bottleneck layers
- 30 Epoch Ran to see “peak” performance
- Heuristics
  - loss/accuracy
  - Confusion matrix
  - Grad Cam

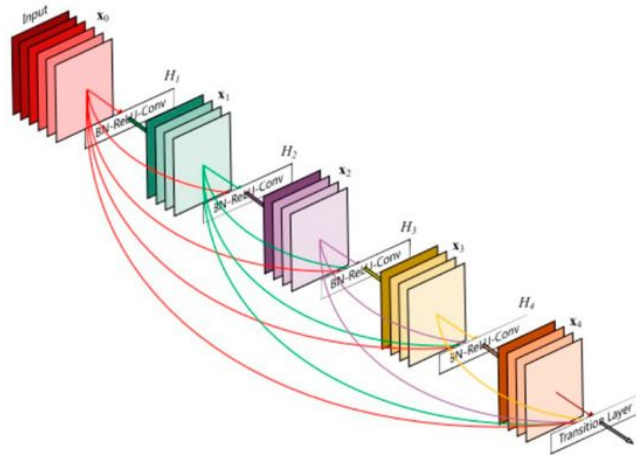
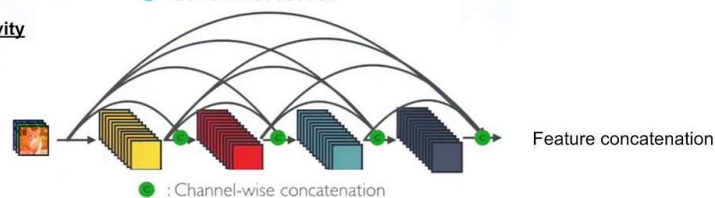
Standard Connectivity



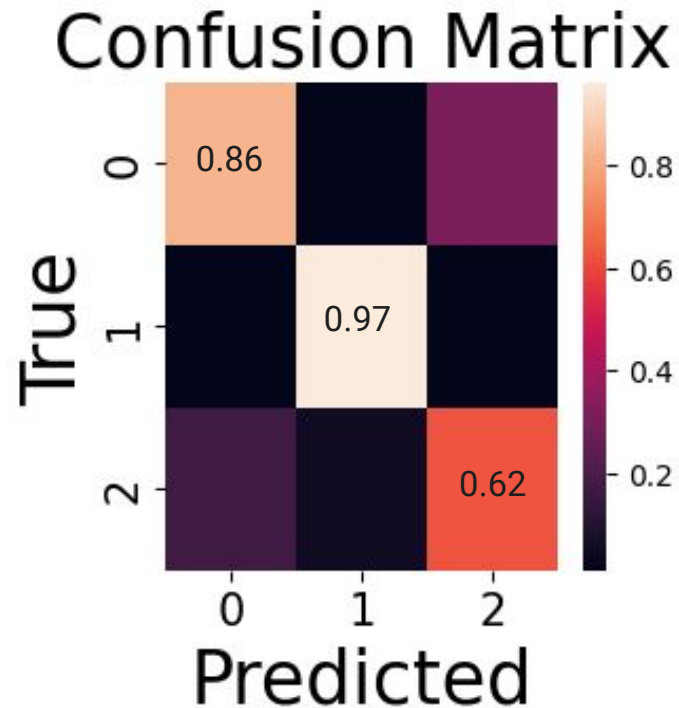
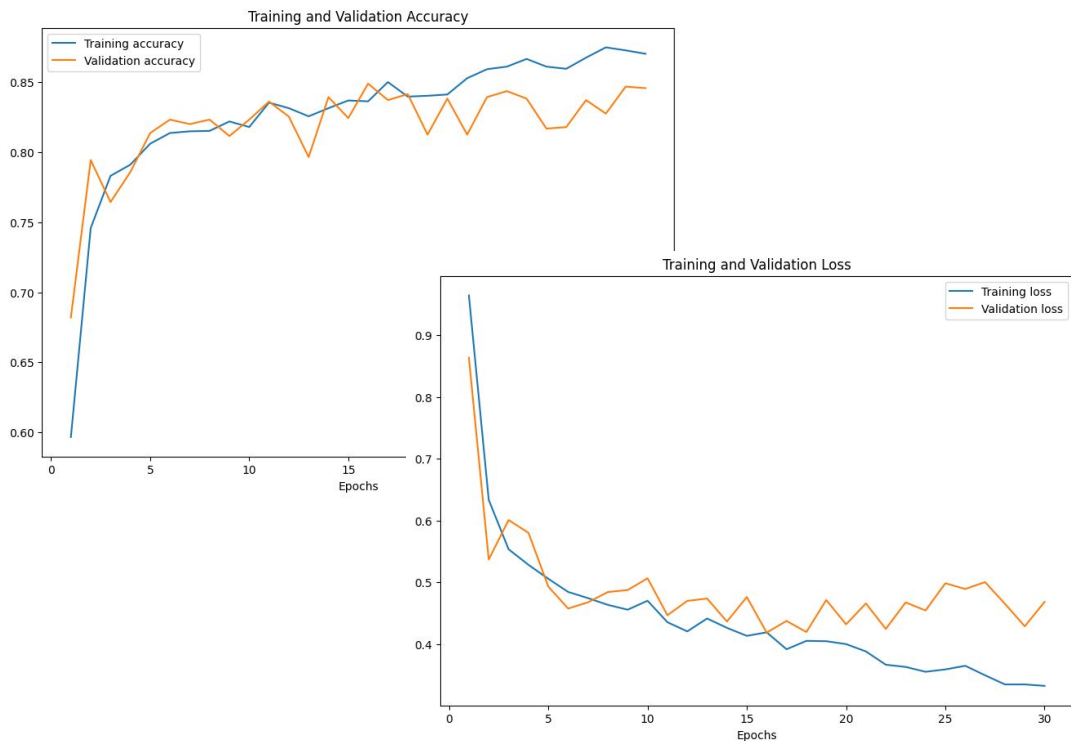
Resnet Connectivity



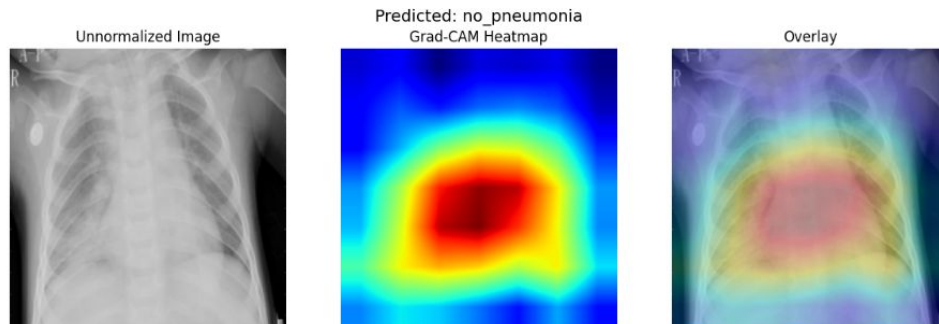
DenseNet Connectivity



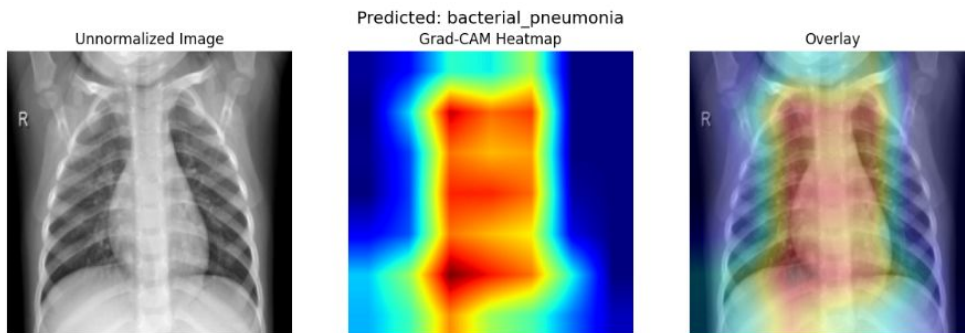
# Key Experiments: DenseNet-121 Results



# Key Experiments: DenseNet-121 Results (cont)



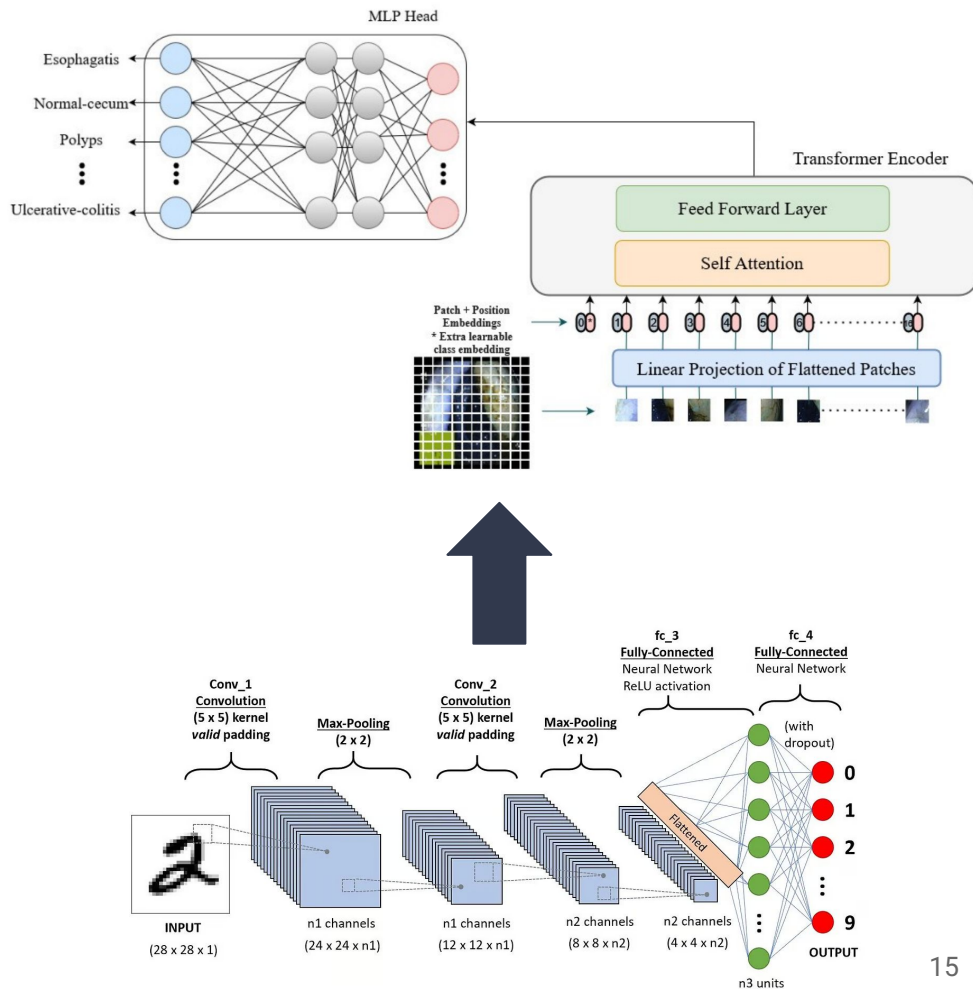
True Class: viral\_pneumonia  
Generating Grad-CAM @features.denseblock4.denselayer16.conv2  
#0: viral\_pneumonia (0.00000)



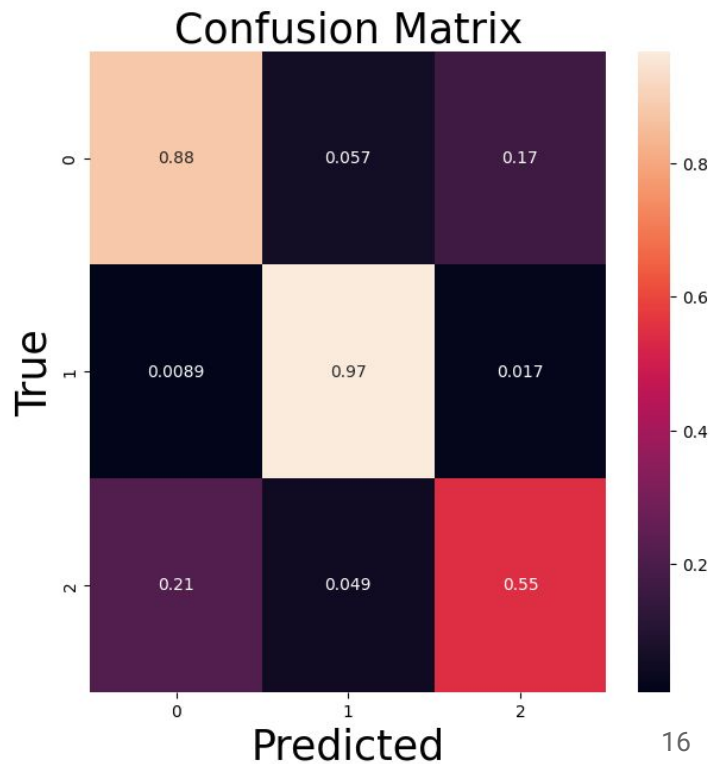
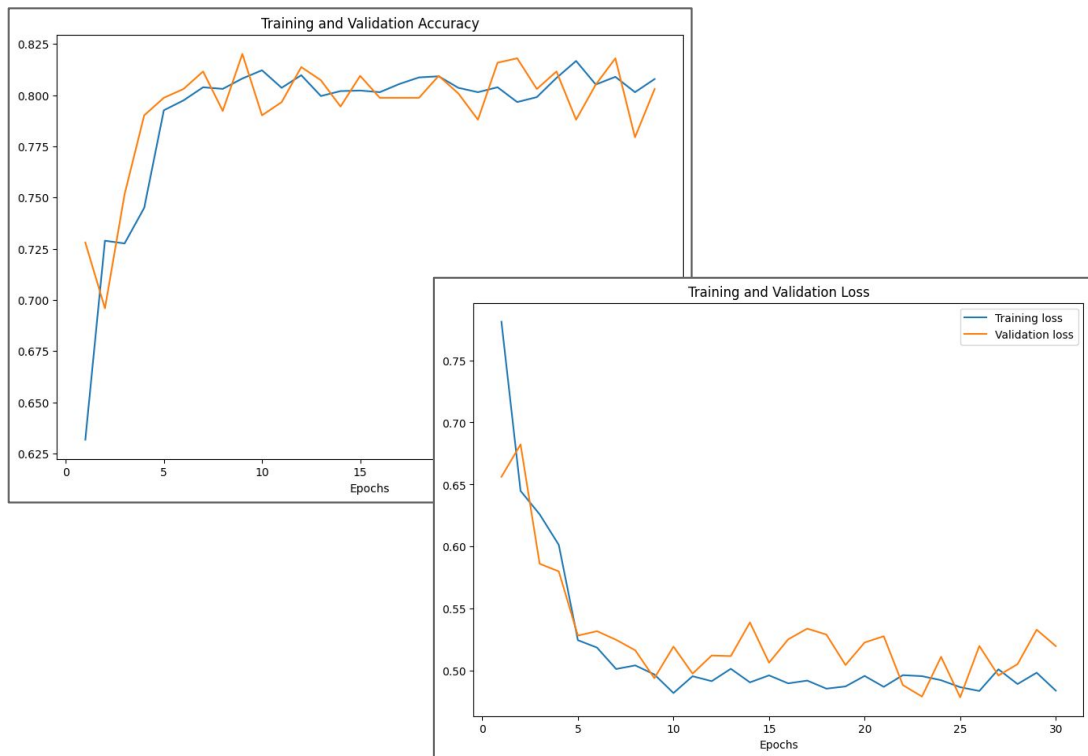
Generating Grad-CAM @features.denseblock4.denselayer16.conv2  
#0: viral\_pneumonia (0.00272)

# Key Experiments: DEiT Model

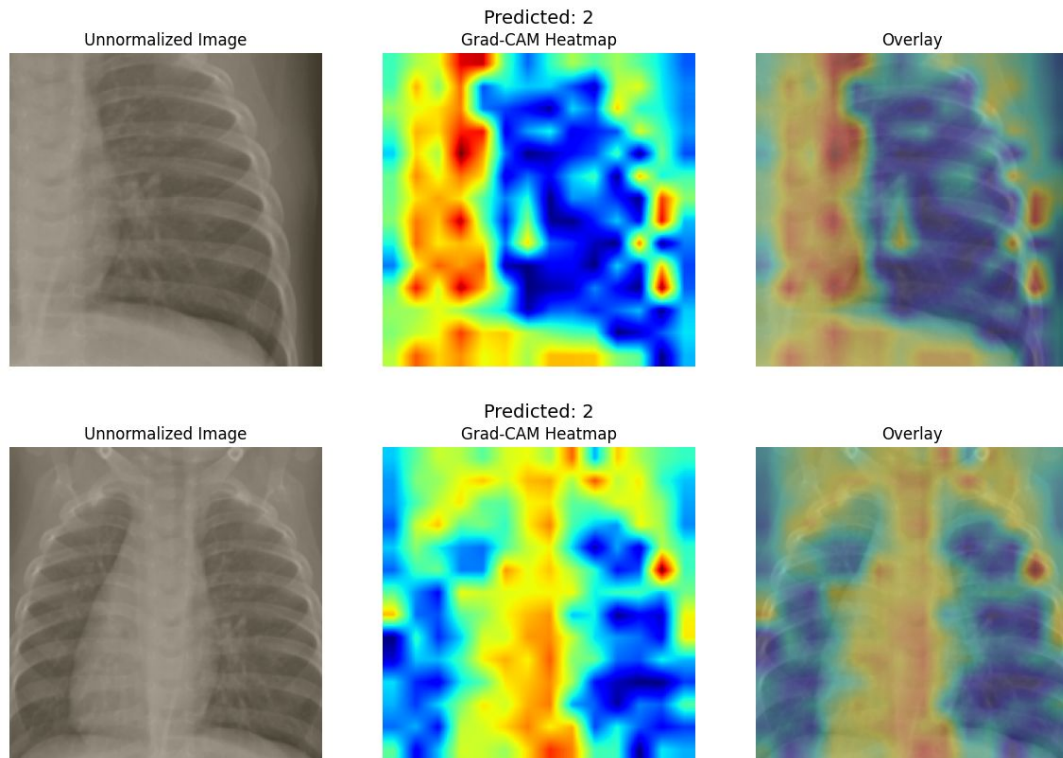
- Refined Model:
  - DEiT-base-16-patch-224
  - DEiT-small & DEiT-tiny\*\*
- Architecture
  - Patch & Position Embeddings
  - Transformer Blocks
    - Multi-head self attention
    - Feed Forward Layer
  - Multi-layer perceptron head
- 30 Epoch Ran
- Heuristics



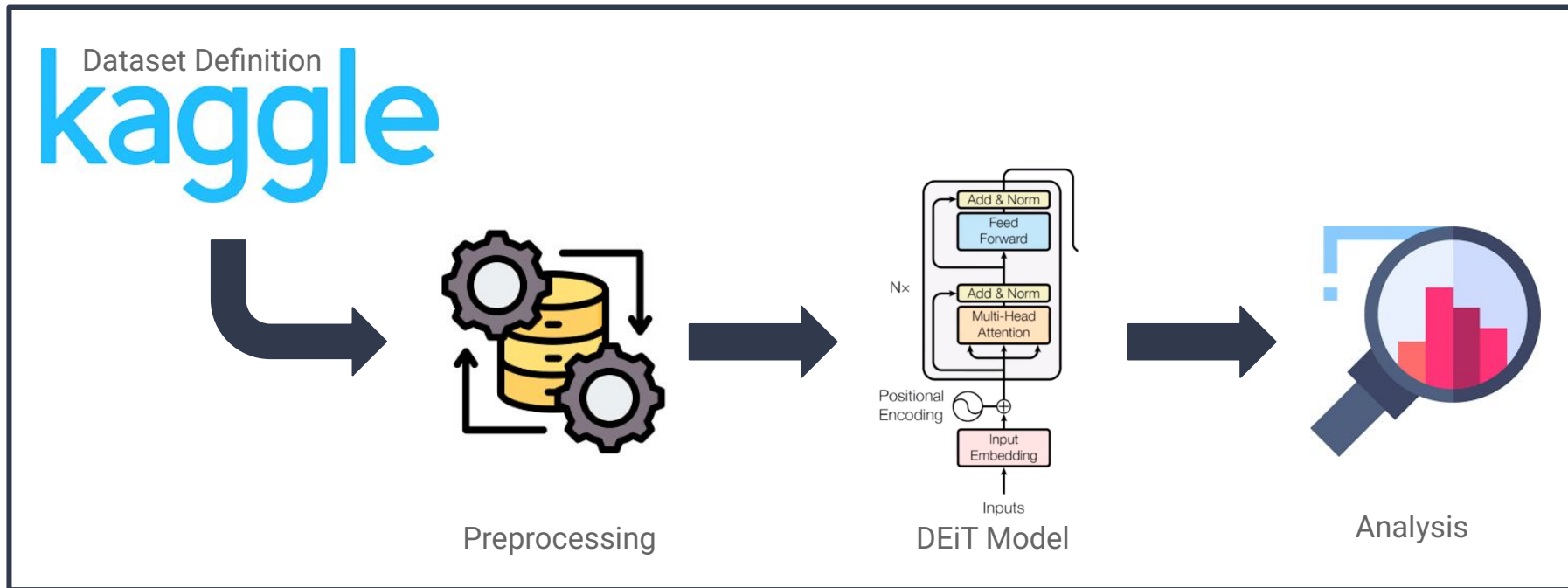
# Key Experiments: DEiT Model Results



# Key Experiments: DEiT Results (cont)

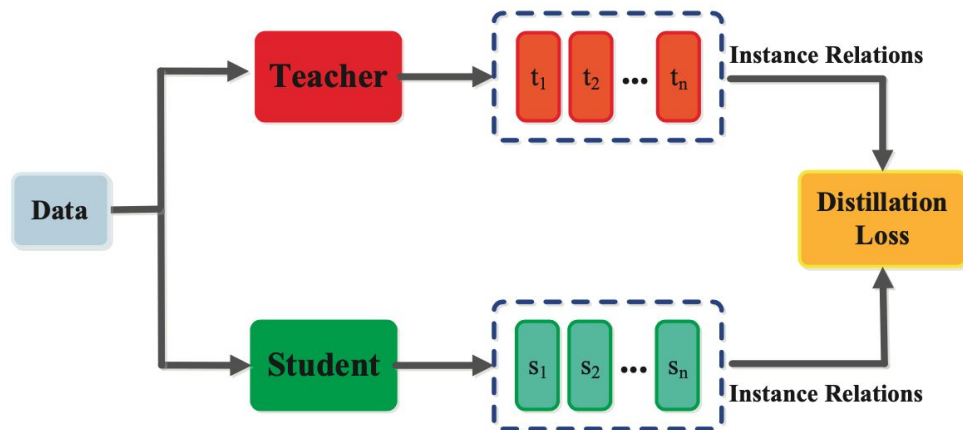
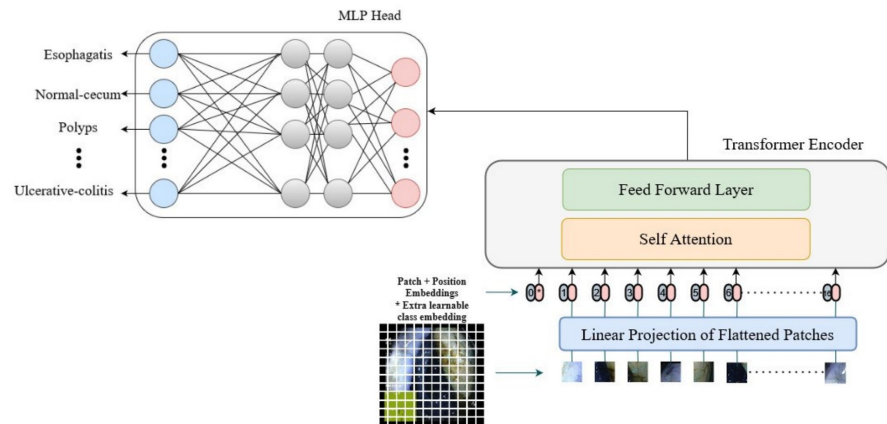


# Final Modeling Pipeline

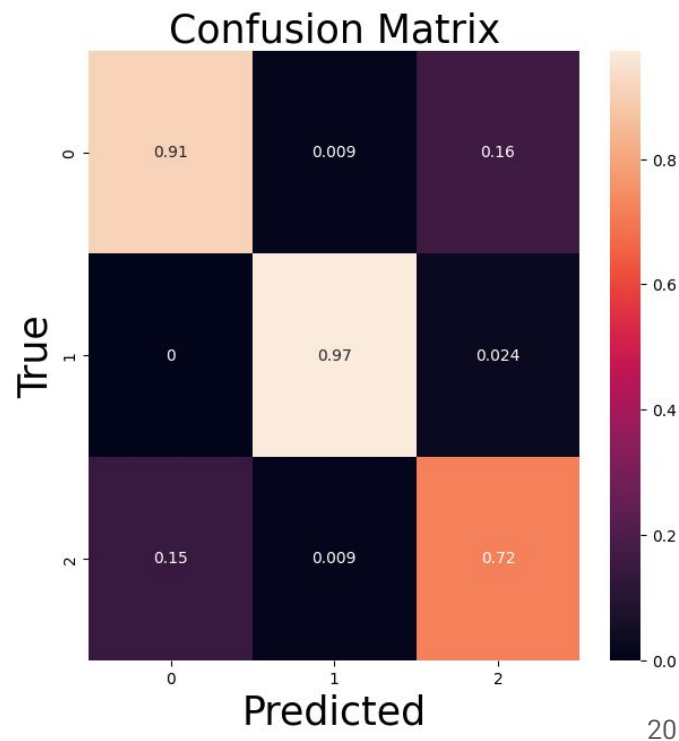
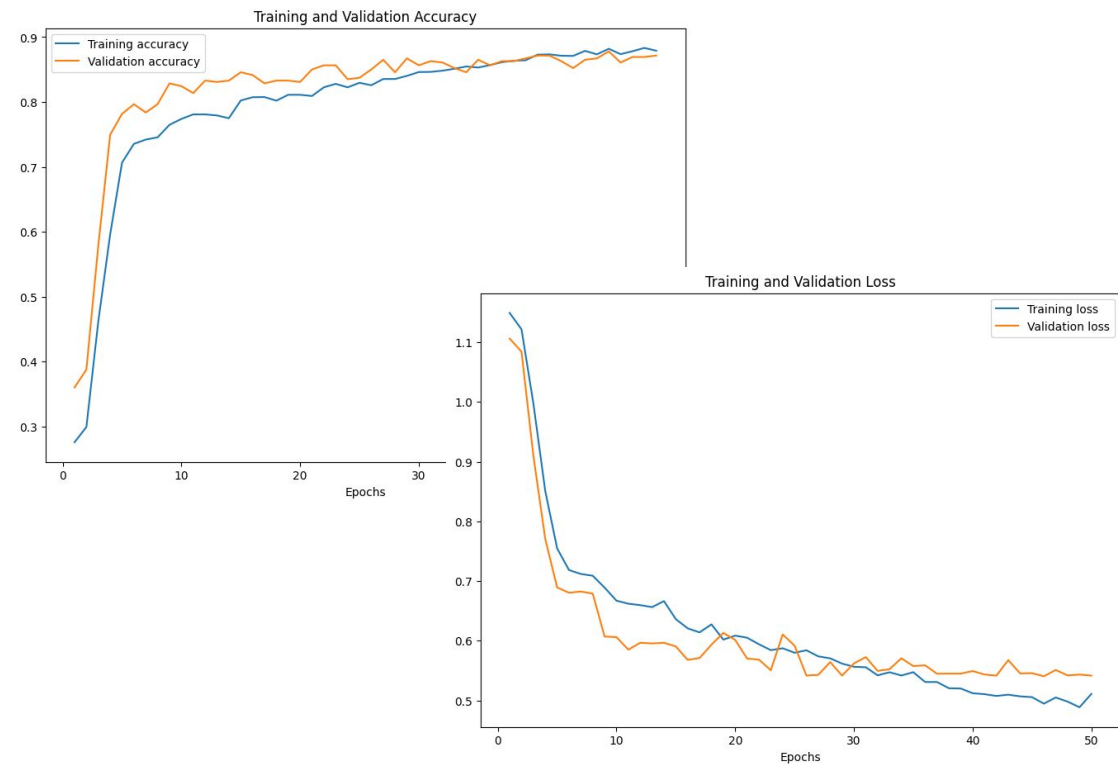


# Final Modeling Pipeline – Solution Model

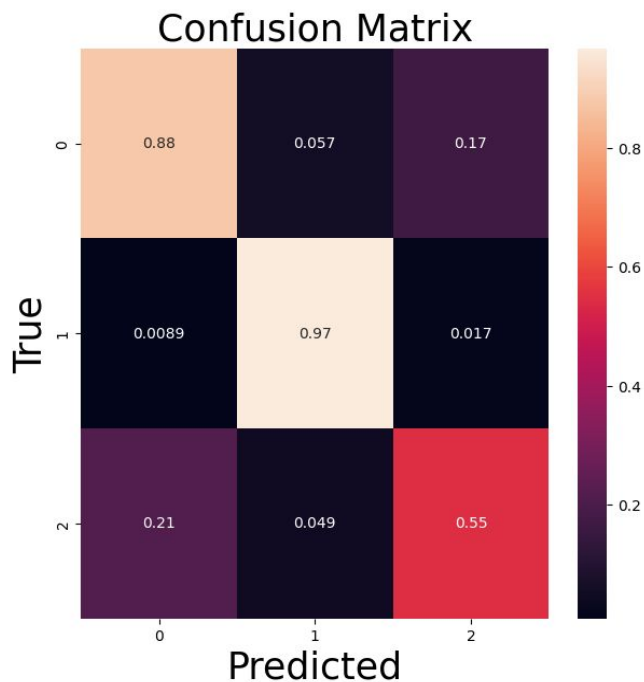
- Solution Model:
  - DEiT-base-distilled-16-patch-224
- Architecture
  - Advanced Deep Learning technique: distillation
  - **Loss function** - Cross Entropy Loss
  - **Optimizer** - AdamW
  - **Learning Rate** - 5e-5
  - **LRScheduler** - Cosine Scheduler



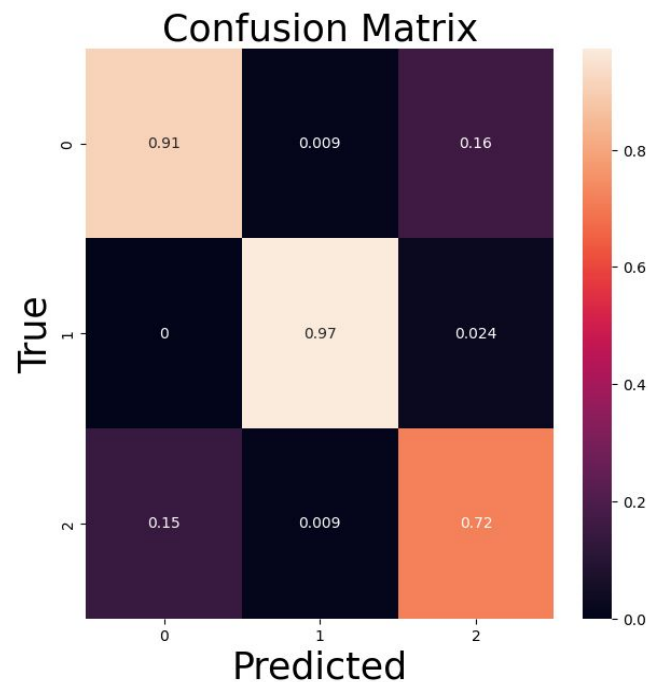
# Final Modeling Pipeline – Solution Model



# Final Modeling Pipeline – Comparison



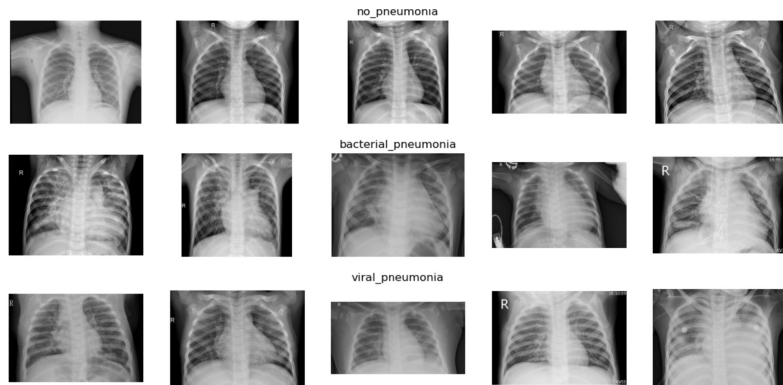
DEiT Base - refined Model



DEiT distilled - Solution Model

# Reflections

- **Hypothesis:** Vision Transformers would be best suited to handle Pneumonia Classifications
  - Not entirely Correct...
- Vision transformers, data preparation pipeline, data analysis
- Data is the most impactful
- Much more work to be done:
  - Viral vs. normal
  - Amount of data



# Contributions

- **Data Visualization:** Steven & Teshy
- **ResNet-18:** Teshy
- **DenseNet-121:** Ethan
- **DEiT:** Steven
- **Solution Model:** Steven & Ethan
- **GradCAM:** Teshy



# Thank You!

Questions?